WORKED SOLUTION

STEP I 2009, Question 9

Mechanics • Projectile motion

STEP I 2009, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

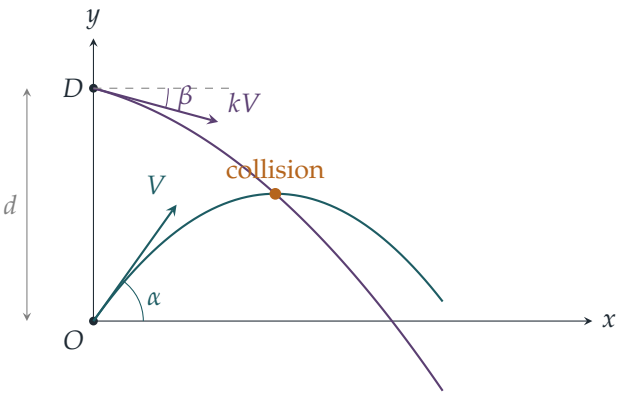
step.maths.org/questions/09-s1-q9

The key idea

Two particles under gravity alone have the *same* acceleration, so the $-\frac{1}{2}gt^2$ terms cancel when we compare their positions: relative to each other, they move in straight lines at constant velocity. Collision means equal horizontal and equal vertical displacements at the same time T. The horizontal equation gives the cosine relation at once. The vertical one involves β, which we remove using $\sin^2\beta = 1 - \cos^2\beta$ and the cosine relation.

Solution

We model P and Q as particles moving freely under gravity, with no air resistance. For them to collide they must be projected in the same vertical plane and in the same horizontal direction. Take O as origin, x horizontal in that direction and y vertically upwards, so D is $(0, d)$.



At time t the positions are

$$\begin{aligned} P: \quad & x = Vt\cos\alpha, && y = Vt\sin\alpha - \tfrac{1}{2}gt^2, \\ Q: \quad & x = kVt\cos\beta, && y = d - kVt\sin\beta - \tfrac{1}{2}gt^2, \end{aligned}$$

where the minus sign in Q's vertical velocity is because β is measured *below* the horizontal.

The cosine relation and the equation for T

At the collision, $t = T > 0$ and the coordinates agree. Horizontally,

$$VT\cos\alpha = kVT\cos\beta,$$

and dividing by $VT \neq 0$ gives $\cos\alpha = k\cos\beta$, as required.

Vertically, the $-\frac{1}{2}gT^2$ terms cancel from both sides, leaving

$$VT\sin\alpha = d - kVT\sin\beta, \qquad \text{so} \qquad kVT\sin\beta = d - VT\sin\alpha.$$

Ask yourself How do we get rid of β?

We know $k\cos\beta$ in terms of α, so we square to bring in $k^2\sin^2\beta$, which is

$$k^2\sin^2\beta = k^2 - k^2\cos^2\beta = k^2 - \cos^2\alpha.$$

Squaring the vertical equation and substituting,

$$\begin{aligned} (k^2 - \cos^2\alpha)V^2T^2 &= d^2 - 2dVT\sin\alpha + V^2T^2\sin^2\alpha, \\ (k^2 - \cos^2\alpha - \sin^2\alpha)V^2T^2 + 2dVT\sin\alpha - d^2 &= 0, \\ (k^2 - 1)V^2T^2 + 2dVT\sin\alpha - d^2 &= 0, \end{aligned}$$

using $\sin^2\alpha + \cos^2\alpha = 1$. This is the required equation.

As a quadratic in T, its roots have product $-d^2/((k^2-1)V^2) < 0$ because $k > 1$. So exactly one root is positive, and that root is the collision time.

Collision at P's highest point

P's vertical velocity is $V\sin\alpha - gt$, which is zero when $t = V\sin\alpha/g$. So here $T = V\sin\alpha/g$. Substituting into the quadratic,

$$(k^2-1)\frac{V^4\sin^2\alpha}{g^2} + \frac{2dV^2\sin^2\alpha}{g} - d^2 = 0.$$

Multiplying by g^2 and collecting the $\sin^2\alpha$ terms gives $\sin^2\alpha\,((k^2-1)V^4 + 2gdV^2) = g^2d^2$, so

$$\boxed{\sin^2\alpha = \frac{g^2d^2}{V^2((k^2-1)V^2 + 2gd)}}$$

The bracket is positive since $k > 1$, so this is well defined.

The inequality

Since $\sin^2\alpha \leq 1$ and the denominator is positive,

$$\begin{aligned} g^2d^2 &\leq (k^2-1)V^4 + 2gdV^2, \\ g^2d^2 - 2gdV^2 + V^4 &\leq k^2V^4, \\ (gd - V^2)^2 &\leq (kV^2)^2. \end{aligned}$$

Both $|gd - V^2|$ and kV^2 are non-negative, so taking square roots preserves the inequality: $|gd - V^2| \leq kV^2$. In particular $gd - V^2 \leq kV^2$, that is,

$$gd \leq (1+k)V^2,$$

as required. Equality would need $\sin^2\alpha = 1$, i.e. P projected vertically.

Checks

- Take $g = 10$, $V = 10$, $k = 2$, $d = 10$. The formula gives $\sin^2\alpha = \frac{100\cdot100}{100(300+200)} = \frac{1}{5}$, so $T = V\sin\alpha/g = 1/\sqrt{5}$. Then the quadratic is $3\cdot100\cdot\frac{1}{5} + 2\cdot10\cdot10\cdot\frac{1}{5} - 100 = 60 + 40 - 100 = 0$. With $\cos\beta = \cos\alpha/2$, both particles are at height 1 at time T. Here $gd = 100 \leq 300 = (1+k)V^2$.
- A numerical check stepped both motions forward for thousands of random V, k, α, T, and confirmed the collision, the cosine relation, the quadratic and the formula for $\sin^2\alpha$ to rounding error.
- Limiting case: if $\alpha \to \frac{1}{2}\pi$ then $\cos\beta \to 0$ too, so both move vertically. P rises to height $V^2/(2g)$ in time V/g, while Q falls by $kV^2/g + V^2/(2g)$, so $d = (1+k)V^2/g$: equality, as predicted.

Common slips

- Getting the sign of Q's vertical velocity wrong: projected *below* the horizontal, it is $-kV\sin\beta$.
- Keeping the $\frac{1}{2}gT^2$ terms: they cancel, and the equation for T has no g in it.
- Taking square roots of $(gd - V^2)^2 \leq k^2V^4$ as $gd - V^2 \leq kV^2$ without noting the modulus. The conclusion is right, but the reason is that $x \leq |x|$.

Beyond STEP

The cancellation of gravity is a first taste of changing frames of reference. In a frame falling freely with acceleration g, both particles move in straight lines at constant speed, and collision problems reduce to geometry of relative velocity. This is the idea behind Einstein's equivalence principle, and it makes many projectile-interception problems

one-liners. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops relative motion and non-inertial frames carefully.

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